Kidney Dysfunction in Heart Failure: Core Curriculum 2025

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The pathophysiology of heart failure (HF) with kidney dysfunction is represented by several maladaptive bidirectional pathways wherein acute or chronic dysfunction of one organ drives acute or chronic dysfunction in the other organ. Suboptimal decongestion, diuretic resistance, and low use rates of guideline-directed medical therapy in individuals with kidney dysfunction and HF contribute to poor cardiovascular and kidney outcomes. Recent developments with the early identification and treatment of diuretic resistance may help mitigate the harmful effects of persistent congestion in individuals with HF. Several classes of guideline-directed medical therapies in HF have been shown to reduce death, hospitalizations for HF, and kidney function decline and improve quality of life. A combination of efficient and personalized strategies to achieve decongestion while optimizing the implementation of evidence-based therapies that modify the trajectory of HF are essential in tandem to reduce adverse outcomes and premature death in this high-risk patient population.

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The Core Curriculum aims to give trainees in nephrology a strong knowledge base in core topics in the specialty by providing an overview of the topic and citing key references, including the foundational literature that led to current clinical approaches.

Introduction

When the heart fails, kidney function frequently worsens. Maladaptive bidirectional cross-talk between the failing heart and kidneys may then ensue, wherein acute or chronic dysfunction of one organ drives acute or chronic dysfunction in the other organ. These pathophysiological pathways extend across hemodynamic, neurohormonal, and inflammatory axes. Suboptimal decongestion, diuretic resistance, and low use rates of guideline-directed medical therapy (GDMT) in individuals with kidney dysfunction and heart failure (HF) contribute to unacceptably high rates of death, hospitalizations for HF, decline in kidney function, and poor quality of life. In this installment of *AJKD*'s Core Curriculum in Nephrology, we explore commonly implicated pathophysiological mechanisms in kidney dysfunction in HF and discuss the principles behind effective decongestion and assessment of adequacy of natriuresis. We also discuss the interpretation of the changes in kidney function in response to decongestion and the use of GDMTs, including strategies to mitigate related adverse effects, especially in the setting of impaired kidney function.

Worsening Kidney Function in HF

Case 1: *A 48-year-old woman with a history of coronary artery disease, hypertension, and type 2 diabetes is hospitalized with progressive dyspnea and leg swelling during the past month. Her medications include aspirin, rosuvastatin, metformin, hydrochlorothiazide, and losartan. Her temperature is 36.7°C, pulse is 105 bpm, respiration rate is 24 breaths/min, and blood pressure is 160/90 mm Hg. Physical examination discloses jugular venous distension, bilateral crackles in the lung bases, and a soft systolic murmur loudest over the cardiac apex. There is grade 2 bipedal pitting edema with warm extremities. Initial laboratory studies show a serum sodium concentration of 133 mEq/L, potassium concentration of 4.0 mEq/L, chloride concentration of 90 mEq/L, serum creatinine concentration of 1.7 mg/dL (baseline, 0.8 mg/dL), serum urea nitrogen (SUN) concentration of 40 mg/dL, and N-terminal pro–brain natriuretic peptide (NT-proBNP) concentration of 3,000 pg/mL. Dipstick urine testing shows trace albumin, and 1-2 red blood cells per high-power field are noted on microscopy, with a urine specific gravity of 1.020. Echocardiography shows a left ventricular ejection fraction of 55% and no pericardial effusion.*

Question 1: What is the major mechanism behind this patient's worsening kidney function?

(a) Kidney venous congestion
(b) Low cardiac output
(c) Prerenal azotemia
(d) Acute tubular injury
(e) Use of an angiotensin receptor blocker (ARB)

For the answer to this question, see the following text.

This individual has acute HF. The conceptual understanding of worsening kidney function in the setting of acute HF has historically centered clinically around reduced cardiac output (ie, forward flow) driving kidney hypoperfusion, with resultant neurohormonal activation,

specifically of the renin-angiotensin-aldosterone system (RAAS) and sympathetic nervous system (SNS), leading to a vicious cycle of sodium avidity and worsening congestion. Low cardiac output is not the dominant mechanism of worsening kidney function in acute HF in a majority of patients and drives kidney dysfunction in a small subset of patients with impending or overt cardiogenic shock.

In this individual, the blood pressure is high and there is clinical evidence of preserved perfusion. Increased right-sided venous filling pressure (and, by extension, kidney venous pressure) is a major determinant of worsening kidney function in HF across the ejection fraction spectrum; therefore, (a) is the best answer to Question 1. This has been demonstrated in animal models of kidney venous clamping and human studies using invasive hemodynamic parameters in which increased right atrial pressure is associated with worsening kidney function in patients with HF independent of cardiac output. It is helpful to conceive the net effect of cardiac output and right-sided venous pressures on kidney perfusion using the construct of "kidney perfusion pressure," which is quantified clinically as the difference between mean arterial pressure and central venous pressure (ideally >60 mm Hg), with disruption of either of these elements critically reducing kidney perfusion in HF.

Mechanisms mediating worsening kidney function in patients with venous hypertension include increased kidney interstitial pressure, increased intratubular pressure, and RAAS/SNS activation, which increases sodium reabsorption, thereby perpetuating congestion. Kidney venous hypertension increases hydrostatic pressures comparably in the peritubular capillaries as well as in the interstitium, resulting in a net even effect with respect to Starling forces between these compartments. However, increased kidney interstitial pressure enhances lymphatic outflow, promoting washout of proteins, significantly reducing colloidal osmotic pressure in the kidney interstitium, which further facilitates sodium reabsorption. In parallel, colloidal osmotic pressure increases in the peritubular capillaries because of the increased filtration fraction in the nephron to maintain the glomerular filtration rate (GFR) in the setting of reduced kidney blood flow (the filtration fraction is the percentage of blood filtered by the glomerulus; GFR divided by renal plasma flow). The resultant increase in sodium and water absorption in the proximal convoluted tubule reduces distal delivery of sodium in HF, which has important therapeutic implications given the sites of action for loop and thiazide diuretic agents. In states of reduced kidney blood flow (due to declining cardiac output), GFR can be preserved initially by a compensatory increase in filtration fraction from efferent arteriolar constriction as a result of increased renin levels in acute HF. When there has been frank and persistent reduction in cardiac output, glomerular filtration is reduced, driven by angiotensin II–mediated constriction of the afferent and efferent arterioles.

Apart from mean arterial pressure and central venous pressure, other mechanisms have been identified that contribute to kidney dysfunction. Increased intraabdominal pressures can lead to compression of renal veins and ureters, leading to overall reductions in glomerular filtration similar to the mechanisms of kidney venous congestion described above. There are emerging data on the role of the kidney's capsule in causing "kidney tamponade," thereby contributing to reduced kidney perfusion pressure. Maldistribution of volume in the splanchnic circulation, which is mobilized in the setting of venoconstriction in HF (a concept called stress volume), also contributes to increased filling pressures. Endothelial dysfunction and inflammatory pathways (including via alterations in the gut microbiome) may also contribute to worsening kidney function in HF. Abnormalities in the microvasculature including reduced nitric oxide and oxidative stress can lead to endothelial damage, promoting downstream inflammation with eventual endothelial-to-mesenchymal transition and fibrosis (Fig 1).

Maladaptive RAAS and SNS activity is upstream and central to these cardiac–kidney interactions and generates a persistent sodium-avid state in patients with HF. This underscores the importance of addressing not only decongestion and its challenges (ie, diuretic resistance) in HF, but the early need to optimize the neurohormonal system in HF as a disease-modifying target, including and especially in individuals with concomitant kidney dysfunction.

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Determining Diuretic Response

Case 2: *A 56-year-old man with a history of HF with reduced ejection fraction (HFrEF) with an EF of 30%, atrial*

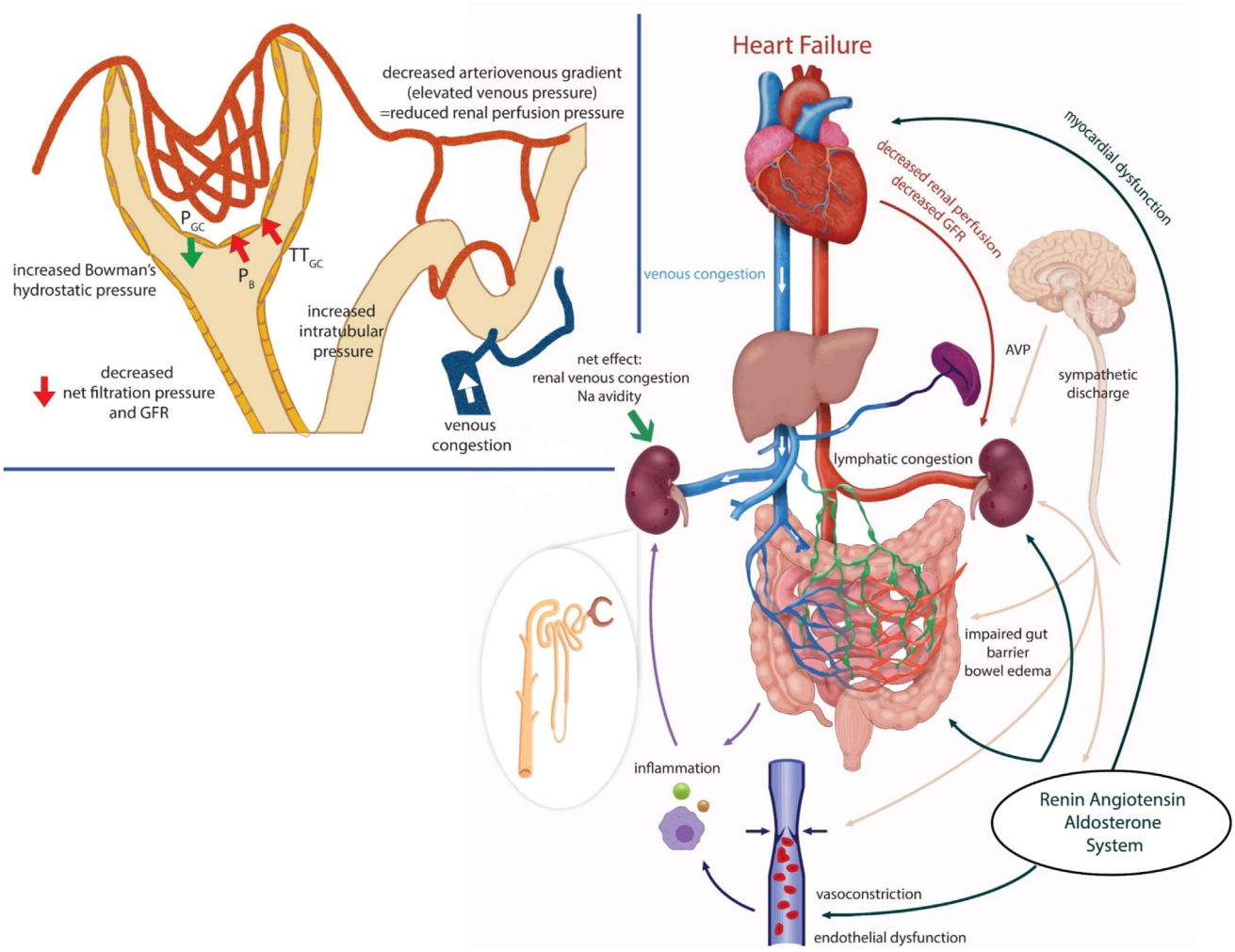


Figure 1. Summary of pathophysiology of cardiac/kidney interactions. Venous congestion plays a major role in cardiac/kidney interactions. Increased venous congestion contributes to a decreased arteriovenous pressure gradient, leading to reduced kidney perfusion pressure. This increased venous pressure is also transmitted across the tubules, leading to increased intratubular pressure, which results in a decreased hydrostatic pressure gradient across the Bowman's capsule, with a net effect of reduced single-nephron glomerular filtration rate. Increased venous pressure also overwhelms the compensatory increase in lymphatic outflow in the kidneys. Finally, the neurohormonal system (renin-angiotensin-aldosterone system) and sympathetic nervous system play central roles in causing and maintaining sodium avidity, thereby worsening the vicious cycle of congestion. Abbreviations: AVP, arginine vasopressin; GFR, glomerular filtration rate; PB, bowman's space hydrostatic pressure; PGC, glomerular capillary hydrostatic pressure; TTGC, glomerular capillary oncotic pressure.

fibrillation, and hypertension is hospitalized with acute HF after a similar presentation 3 weeks earlier. His discharge medications include valsartan, carvedilol, spironolactone, apixaban, and furosemide (40 mg/d orally). His temperature is 36°C, pulse is 110 bpm, respiration rate is 24 breaths/min, and blood pressure is 120/80 mm Hg. Physical examination discloses jugular venous distention, bilateral crackles in the lung bases, and an irregularly irregular rhythm. There is grade 2 pitting edema below the knees bilaterally. Initial laboratory studies show a sodium concentration of 135 mEq/L, potassium concentration of 3.6 mEq/L, chloride concentration of 92 mEq/L, serum creatinine concentration of 1.3 mg/dL (baseline, 0.9 mg/dL), SUN concentration of 15 mg/dL, and NT-proBNP concentration of 3,500 pg/mL (discharge NT-proBNP, 900 pg/mL).

Question 2: What is the most appropriate starting dose for diuretic agents in this patient on admission?

(a) Furosemide 40 mg intravenously
(b) Furosemide 80 mg orally
(c) Bumetanide 1 mg intravenously
(d) Torsemide 40 mg orally
(e) Furosemide 100 mg intravenously

For the answer to this question, see the following text.

Most societal HF guidelines have recommended to start intravenous loop diuretic therapy with at least twice the daily home dose for treatment of acute HF. Most of the early evidence with loop diuretic agent use is from the

DOSE trial, which was a randomized trial of patients hospitalized with acute HF. The study compared a high-dose (2.5× the home oral dose) versus a low-dose diuretic approach. Although the trial showed only a nonsignificant trend toward improvement in patients' global assessment of symptoms, it did show some improvement in secondary outcomes such as net fluid loss and weight change. The DOSE trial also demonstrated no differences between continuous intravenous loop diuretic agent infusion and bolus intermittent dosing. During an acute exacerbation of HF, it is important to use adequate doses of loop diuretic agents and the proper route to be able to relieve congestion. Loop diuretic agents have steep dose–response curves. When they have reached a level threshold of maximal activity of their diuretic effect, further increases in dose would not necessarily increase the natriuretic effects. Further administration of diuretic doses, however, can keep the drug levels higher than this ceiling threshold, allowing for more time for the diuretic agents to exert their effects. Patients with acute HF tend to have variable degrees of intestinal edema, leading to unpredictable absorption of oral diuretic agents regardless of their innate bioavailability. In patients with HF in whom the maladaptive heightened RAAS/SNS activity leads to persistent sodium avidity, a higher dose of diuretic agents is needed to reach the threshold of diuretic effect. This makes the intravenous route preferred; therefore, option (e) is the best answer to Question 2.

There are pharmacochemical differences between loop diuretic agents, with torsemide and bumetanide exhibiting greater oral bioavailability and preclinical/clinical data supporting favorable modulation of the RAAS with possible underlying disease modification effects in HF for torsemide. However, the TRANSFORM-HF trial failed to demonstrate a significant difference in all-cause mortality between torsemide and furosemide in individuals discharged after an acute HF hospitalization. A dose of 40 mg of intravenous furosemide is equivalent to 1 mg of bumetanide or 20 mg of torsemide. However, recent data from TRANSFORM suggest a 4:1 ratio between furosemide and torsemide doses as being equally natriuretic. The other doses of diuretic agents in the question above would be insufficient for an adequate diuretic response.

Outpatient intravenous administration of loop diuretic therapy such as in an infusion center or observation/HF care unit is a safe and feasible alternative strategy compared with the same treatment in the hospitalized setting. Subcutaneous furosemide administration is also a recently available alternate approach to bypass suboptimal diuretic responses to oral diuretic agents.

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CASE 2, CONTINUED

Question 3: What is the best way to accurately assess diuretic response during decongestion for hospitalized HF?

(a) Daily measurement of patient weight
(b) Urine sodium concentration measured 2 hours after diuretic administration
(c) Charted 24-hour urine output
(d) Clinical signs and symptoms assessed at the bedside
(e) Trend in serum urea nitrogen and creatinine levels

For the answer to this question, see the following text.

Daily weights and urine output measurements in acute HF are affected by myriad patient- and systems-related factors, making them relatively unreliable in measuring diuretic response adequacy. Furthermore, these measures take time (≥1 day for weight change) to be able to observe a trend or a significant change. Clinical signs and symptoms are difficult to accurately ascertain. As mentioned above, acute HF is a highly sodium-avid state driven by excess and maladaptive neurohormonal responses (via the RAAS and SNS) to the failing heart. Therefore, the degree of sodium avidity can potentially reflect the degree of diuretic resistance and the efficacy of natriuresis achieved by diuretic therapy and quickly help in ascertaining necessary changes for diuretic agent dosing; therefore, (b) is the best answer to Question 3. The use of a spot urine sodium measurement 2 hours after the first diuretic dose is a reliable way of predicting subsequent 6-hour natriuresis, as validated by the natriuretic response prediction equation. A spot urine sodium concentration <50-70 mEq/L at 2 hours after loop diuretic agent administration or an hourly urine output <100-150 mL during the first 6 hours denotes an insufficient diuretic response. This can allow decision-making for rapid diuretic agent uptitration to improve natriuresis. A single-center prospective pragmatic randomized clinical trial (the PUSH-AHF trial) explored

natriuresis-guided therapy using spot urine sodium values checked at predetermined intervals compared with the standard of care. It showed a higher degree of natriuresis in the natriuresis-guided arm, but no significant differences were seen in clinical end points of all-cause mortality or first rehospitalization for HF. In a larger multicenter nonrandomized study (the ENACT-HF study) comparing a natriuresis-guided diuretic protocol versus standard of care, the natriuresis-guided protocol was associated with greater natriuresis and diuresis and shorter length of stay, without an increase in safety events. It is important to note that most of the studies of natriuresis-guided decongestion have used spot urine sodium measurements as a detector of early (within 24-72 hours) insufficient diuretic response. Use later in the hospitalization is likely to be limited as a result of rapid tubular adaptation.

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Diuretic Resistance and Enhanced Decongestive Strategies

Case 4: *A 53-year-old woman with a history of HF with preserved EF with an EF of 55%, diabetes, hypertension, and chronic kidney disease (CKD; KDIGO [Kidney Disease: Improving Global Outcomes] stage G4 A3) is hospitalized with an HF exacerbation. Outpatient medications include lisinopril, metoprolol succinate, dapagliflozin, and torsemide (80 mg) daily. Blood pressure is 110/70 mm Hg, heart rate is 80 bpm, respiratory rate is 24 breaths/min, and temperature is 37.3°C. Initial laboratory studies show a sodium concentration of 132 mEq/L, potassium concentration of 4.5 mEq/L, chloride concentration of 92 mEq/L, serum creatinine concentration of 3.1 mg/dL (at baseline), SUN concentration of 55 mg/dL, and NT-proBNP concentration of 5,000 pg/mL. The patient undergoes diuresis with intravenous furosemide 200 mg twice daily, but, during the first 24 hours, the net intake/output status is +300 mL. There was no noted weight change in 24 hours. The patient continues to receive 4-6 L of oxygen.*

Question 4: What is the next best step for this patient?

(a) Increase furosemide to 400 mg intravenously twice daily
(b) Change diuretic agent to bumetanide 4 mg intravenously twice daily
(c) Add metolazone 5 mg/d
(d) Discontinue lisinopril treatment
(e) Initiate ultrafiltration

For the answer to this question, see the following text.

This patient has diuretic resistance, defined as the attenuation of the maximal diuretic effect that ultimately limits sodium and chloride excretion. Related to this is the concept of diuretic efficiency, which refers to the net fluid lost/urine output per milligram of loop diuretic agent (usually in per–40-mg furosemide dose equivalents). Diuretic resistance is associated with poor outcomes such as worsening kidney function, recurrent HF hospitalizations, and mortality. The patient has advanced CKD, which, in itself, may contribute to diuretic resistance, even though diuretic resistance is usually not entirely explained by CKD alone. Accumulation of organic anions that compete for diuretic secretion in the proximal convoluted tubule, and overall diminished filtered load of sodium from a decreased GFR from critically reduced nephron mass in more advanced CKD, are mechanisms via which CKD can contribute to diuretic resistance. Notably, this patient is already receiving high doses of an oral diuretic medication. Generally, the natriuretic response to each subsequent dose decreases, constituting a phenomenon called diuretic braking. This results from the combined hyperactivity of the RAAS and SNS. There is also nephron remodeling, particularly distal tubular hypertrophy that increases distal sodium reabsorption and aldosterone-triggered responses in the collecting duct with increased epithelial sodium channel–mediated reabsorption of sodium. This redistribution of sodium delivery and reabsorption is also seen with sodium glucose cotransporter inhibitors, as has been demonstrated elegantly using lithium excretion studies. Hypochloremia and metabolic alkalosis both antagonize the effects of loop diuretic agents by reducing the intraluminal chloride gradient and triggering the adaptive neurohormonal responses described above.

Pendrin is a chloride-bicarbonate anion exchanger located in the collecting ducts that plays a major role in chloride reabsorption from the tubules in exchange for bicarbonate secretion. Pendrin acts in concert with the sodium-dependent chloride and bicarbonate exchanger transport that cotransports sodium and bicarbonate for reabsorption in exchange for chloride secretion. This

pathway serves as another independent pathway for sodium reabsorption that is potentially independent of the epithelial sodium channels or neurohormonal influences that may act as a therapeutic target (eg, using acetazolamide) to inhibit pendrin to counter sodium reabsorption and diuretic resistance (Fig 2).

Various approaches have been attempted, including the concept of sequential nephron blockade to achieve "enhanced decongestion" as an outcome in acute HF. This approach uses different diuretic agents acting across nephron segments to help augment the overall diuretic response and bypass the previously mentioned adaptive mechanisms; therefore, (c) is the best answer to Question 4. Small studies have shown the efficacy of the addition of thiazide-type diuretic agents to help augment the overall diuretic response. However, until recently, combinations of different diuretic agent types were not studied in larger randomized trials. The CLOROTIC trial randomized 230 hospitalized patients with acute HF to receive hydrochlorothiazide or placebo in addition to furosemide standard therapy. Patients receiving hydrochlorothiazide had greater weight loss and volume of diuresis at the expense of greater rates of impaired kidney function. It is important to clarify that the impaired kidney function in this trial is of uncertain clinical significance because no assessment of tubular injury was performed, and the overall clinical outcomes remained neutral. The ADVOR randomized trial examined acetazolamide, a carbonic anhydrase inhibitor that predominantly acts in the proximal convoluted tubule reducing sodium reabsorption versus placebo, in addition to standard loop diuretic therapies in acute HF with varied EFs. This trial randomized 519 patients hospitalized with HF to receive acetazolamide intravenously once daily or placebo in addition to their intravenous loop diuretic regimen. The main end points of successful decongestion (defined by absence of signs of volume overload within 3 days after randomization) was higher in the acetazolamide group than in placebo group (42.2% vs 30.5%). There were also greater rates of natriuresis and urine volume documented in the acetazolamide group. However, no differences between groups were noted for hard clinical end points such as hospitalizations for HF, mortality, or quality of life.

Another proximal convoluted tubule agent, empagliflozin, was studied in acute HF in the EMPULSE (Empagliflozin in Patients Hospitalized for Acute Heart Failure) trial. This trial randomized 530 patients hospitalized with HF (with varying EFs) to receive empagliflozin or placebo in addition to loop diuretic therapy. The primary end point of a hierarchical composite "win ratio" of death, HF exacerbations, and change in symptoms measured by the Kansas City Cardiomyopathy Questionnaire at 90 days showed significant benefit in favor of empagliflozin (win ratio, 1.36; 95% CI, 1.09–1.68). In a separate analysis, empagliflozin also led to significant reductions in markers of decongestion, including weight, weights adjusted for mean daily loop diuretic agent dose, and natriuretic peptide levels, compared with placebo.

Other trials testing add-on therapies such as ATHENA-HF (spironolactone) and EVEREST (tolvaptan) in acute HF did not show benefits in terms of hard clinical outcomes such as mortality; however, tolvaptan did show the benefits of improvements in dyspnea, weight, and edema. Tolvaptan is not necessarily a diuretic agent but more of an aquaretic agent, whose mechanism of action is a vasopressin V2 receptor antagonist leading to predominantly water excretion. There are not enough head-to-head comparative studies comparing these agents used for sequential nephron blockade. In a small randomized trial (3T trial) of patients with diuretic resistance that compared oral metolazone, intravenous chlorothiazide, and tolvaptan added to standard high-dose furosemide, weight loss was similar across all 3 strategies, with improved overall diuretic efficacy (Table 1).

The use of hypertonic saline solution for decongestion in HF, although seemingly counterintuitive, has recently gained traction, with the theoretical approach of disruption of the sodium-avidity signal tied to neurohormonal activation. Some studies have shown an improvement in urine output and diuretic efficiency among patients with acute decompensated HF with administration of hypertonic saline solution with high doses of loop diuretic agents. However, high-quality evidence is still lacking for this approach, but it can be potentially considered in refractory cases, especially in patients who are poor candidates for ultrafiltration.

So far, most trials (including the recent ones emphasizing enhanced decongestion strategies) did not select patients with or at high risk for diuretic resistance, but instead enrolled patients with acute HF in general. Among the contemporary trials, the CLOROTIC trial enrolled patients at higher risk for diuretic resistance, given a baseline diuretic agent use of 80-240 mg furosemide equivalents and an estimated GFR (eGFR) range of 32-58 ml/min/1.73 m^2, and struggled with slow enrollment, reaching a sample size of only 230 patients. To this end, the wider availability and validation of early detection of diuretic resistance using sodium output is helpful in informing patient selection in future trials for the optimization of decongestion strategies.

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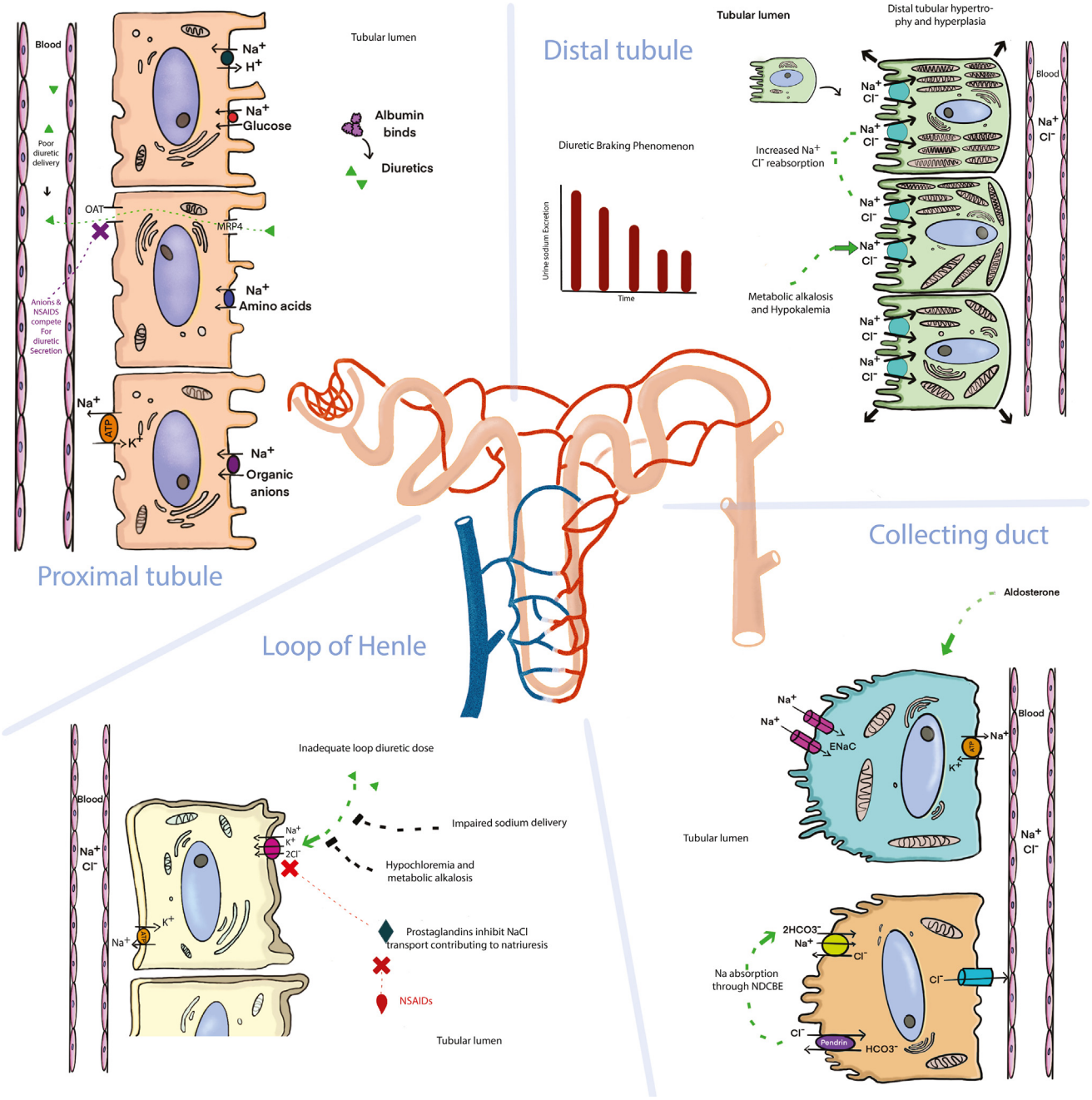


Figure 2. Summary of mechanisms of diuretic resistance. This figure summarizes the mechanisms involved in diuretic resistance based on the site of the nephron involved. The upper left area corresponds to the proximal tubule, where initial poor diuretic delivery from inadequate dosing or poor oral absorption can already be seen. In addition to this, anions and urate, which are increased in cardiorenal disease states as well as nonsteroidal anti-inflammatory drug use, compete with diuretic agents for tubular secretion via the organic anion transporter and are subsequently secreted to the lumen through MRP4 (multidrug resistance–associated protein 4). When secreted, albumin in the tubules can also bind the diuretic agents, further reducing drug availability going to the main site of action in the thick ascending limb. Going to the next segment in the loop of Henle in the lower left corner, the $Na^+K^+2Cl^-$ transporter is the main site of action of loop diuretic agents. This is antagonized by the effects of hypochloremia and metabolic alkalosis by reduction of intraluminal chloride and overall inadequate tubular response to the delivered diuretic agent related to the triggered elevated neurohormonal response. Poor sodium delivery from a diminished filtered load of sodium due to chronic kidney disease can play a part in diuretic resistance. The majority of diuretic resistance is postulated to be mediated in the distal segments of the nephron. Seen in the right side of the diagram, distal tubular hypertrophy and hyperplasia lead to compensatory increases in NaCl reabsorption through the sodium chloride cotransporter, and this is again further enhanced by metabolic alkalosis and hypokalemia. There is also hypertrophy of the principal and intercalated cells in the collecting duct, with increased sodium reabsorption mediated by aldosterone via the epithelial sodium channels, as well as the combined physiologic effects of pendrin and the sodium-dependent chloride/bicarbonate exchanger. The overall net effect, together with the combined sympathetic nervous system and

Table 1. Agents Used for Sequential Nephron Blockade

Site/Agent	Landmark Trial	Relief of Congestion	Clinical Outcomes Benefit
Proximal tubule			
Acetazolamide	ADVOR	Yes	No
SGLT2 inhibitor	EMPULSE	Yes	Yes
Distal tubule			
Thiazide	CLOROTIC	Yes	No
Collecting duct			
MRA	ATHENA-HF	Yes	No
Tolvaptan	EVEREST	Yes	No

Abbreviations: MRA, mineralocorticoid receptor blocker; SGL2, sodium/glucose cotransporter 2.

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Progressive Worsening of Kidney Function and Inadequate Response to Decongestion

Case 5: *A 60-year-old man with a history of HFrEF (EF, 25%), diabetes, hypertension, atrial fibrillation, and CKD is hospitalized with HF. He has not been able to tolerate escalation of GDMTs because of relative hypotension and hyperkalemia. His home medications include furosemide 80 mg twice daily, apixaban, and metoprolol succinate. His temperature is 36.1°C, pulse is 120 bpm, respiration rate is 22 breaths/min, and blood pressure is 90/60 mm Hg while receiving 2 L of oxygen via nasal cannula (O_2 saturation at 94%). Physical examination discloses jugular venous distension, bilateral crackles in the lung bases, and an irregularly irregular heart rhythm. There is grade 2 pitting edema up to the level of the thighs and lower abdomen, with delayed capillary refill at 3-4 seconds. Initial laboratory studies show a sodium concentration of 130 mEq/L, potassium concentration of 4.5 mEq/L, chloride concentration of 89 mEq/L, serum creatinine concentration of 2.4 mg/dL (at baseline), SUN concentration of 60 mg/dL, and NT-proBNP concentration of 5,000 pg/mL. The patient undergoes diuresis with intravenous furosemide 200 mg twice daily, augmented with intravenous chlorothiazide 500 mg. However, after the first 24 hours, urine output was 1,500 mL, which was a net negative intake/output of −300 mL. There was no noted weight change. In the next day, the patient's 24-hour urine output was reduced to 1,000 mL, and his serum creatinine concentration trended up to 3.2 mg/dL, with a serum bicarbonate concentration of 26 mEq/L and a serum lactate level of 4 mmol/L (normal, <2 mmol/L).*

Question 5: What is the next best step for this patient?

(a) Change to furosemide continuous infusion and add acetazolamide
(b) Change furosemide to bumetanide infusion
(c) Add metolazone 30 minutes before furosemide
(d) Arrange for right heart catheterization
(e) Initiate ultrafiltration

For the answer to this question, see the following text.

Fluctuations in markers of glomerular function (serum creatinine and cystatin C) are expected in the context of decongestion and GDMT initiation/titration (especially RAAS inhibitors) and do not reflect intrinsic kidney tubular injury per se. However, recognition of true tubular injury secondary to low cardiac output is necessary, especially in a patient who is intolerant of GDMT. The ESCAPE trial showed no difference in clinical outcomes of mortality in routine monitoring of invasive hemodynamic parameters in patients with acute HF; however, having invasive hemodynamic measurements can be beneficial to guide therapy in certain cases. Worsening kidney function despite appropriate decongestion efforts, inadequate responses to diuretic therapy despite pharmacologic escalation with multiple diuretic agent classes, uncertainties about volume status, or a concern for low cardiac output should prompt early consideration for right heart catheterization; therefore, (d) is the best answer to Question 5. This allows early targeted therapies such as initiation of inotropes or acute mechanical circulatory support to help augment cardiac output in cases of cardiogenic shock, in addition to continued efforts toward decongestion after reestablishing perfusion. Other noninvasive ways to determine the degree of congestion such as point-of-care ultrasound including lung ultrasound, assessment of inferior vena cava caliber/collapsibility, and the Venous Excess Ultrasound score are helpful, but caveats exist with its inherent operator- and organ-specific assumptions and limitations. Evidence is lacking about the efficacy of administration of thiazide diuretic agents 30 minutes before the loop diuretic agent. This practice was adapted from a time when no intravenous thiazide or thiazide-like diuretic formulations were available. In diuretic-refractory HF, initiation of ultrafiltration to aid in decongestion can be considered. The early UNLOAD trial showed greater weight and fluid loss and reduction in 90-day rehospitalization with early ultrafiltration compared with standard intravenous diuretic therapy. However, another trial, CARRESS-HF, showed that the use of early fixed-rate ultrafiltration in patients with HF and worsening kidney function, compared with a stepped pharmacologic decongestive approach, showed no difference in weight loss

Figure 2 (Con'd). renin-angiotensin-aldosterone system activation, is the observed reduction in natriuresis over time with chronic administration of diuretic agents, also known as the diuretic braking phenomenon. Abbreviations: ENaC, epithelial sodium channel; NSAID, nonsteroidal anti-inflammatory drug; OAT, organic anion transporter.

in the setting of higher rates of increase in serum creatinine level and adverse events in the ultrafiltration group mainly driven by infection and active bleeding. Finally, the AVOID trial compared adjustable ultrafiltration (versus fixed-rate ultrafiltration in CARRESS-HF) with adjustable intravenous loop diuretic agents. This study found that the adjustable ultrafiltration group showed a trend toward a longer time to the first HF event, but this study was prematurely terminated by the sponsor and enrolled only 110 patients. Careful patient selection and timing are key in the initiation of stepped ultrafiltration in patients with persistent congestion with HF (algorithms shown in Figs 3 and 4).

Additional Readings

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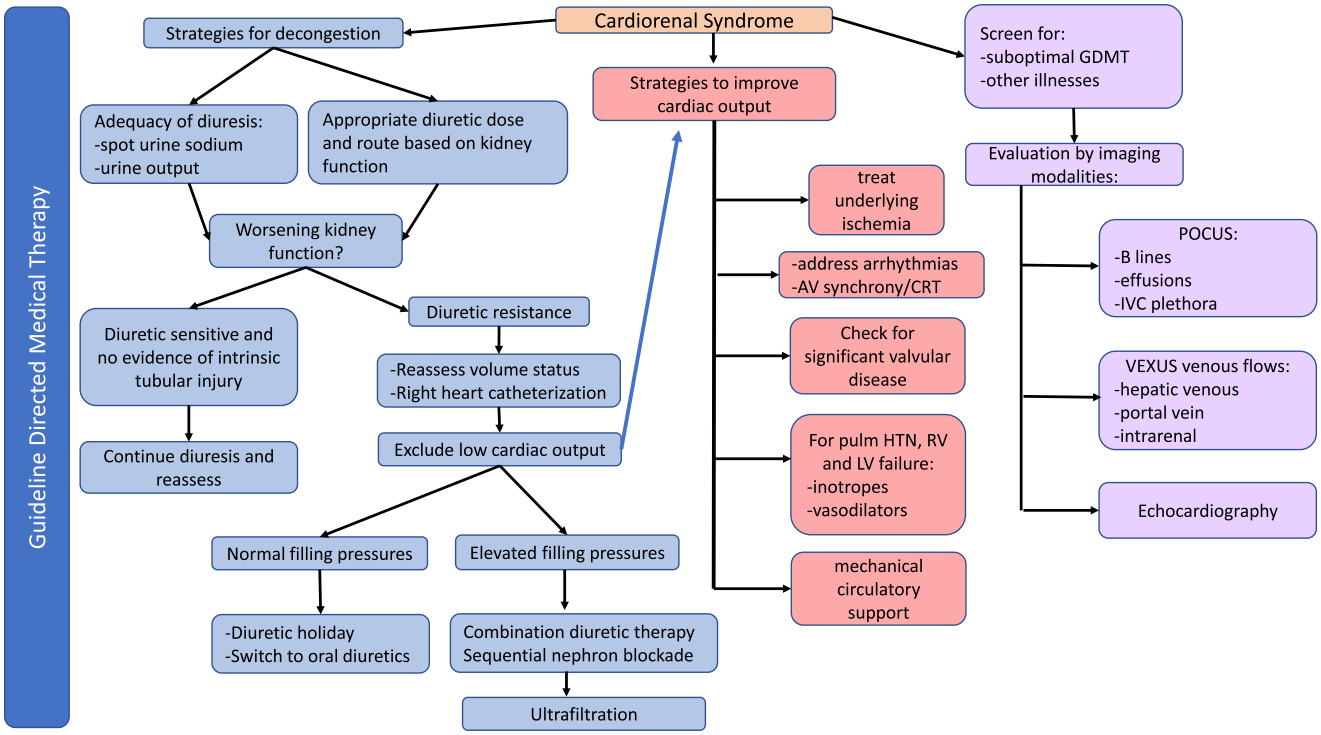


Figure 3. Algorithm for decongestion in the setting of heart failure and kidney dysfunction. When encountering worsening kidney function in the setting of heart failure, it is essential to screen for risk factors such as dietary factors, inadequate guideline-directed medical therapy use, and other intercurrent illnesses as triggers for an acute exacerbation. Assessment of volume status clinically and with the use of supplemental imaging modalities such as point-of-care ultrasound, the Venous Excess Ultrasound score, and echocardiography remains key. If volume overload is deemed to have occurred, a trial of diuresis is given with the appropriate dose according to baseline kidney function and baseline home diuretic doses. Adequacy of diuresis and diuretic resistance can be identified early using the strategy of spot urine sodium measurement 2 hours after diuretic agents are given. A spot urine sodium concentration of <50-70 mEq/L 2 hours after diuretic agent administration or an hourly urine output concentration of <100-150 mL during the first 6 hours denotes an insufficient diuretic response. If the patient's response is appropriate, continued monitoring and reassessment are performed. If diuretic resistance is encountered, repeat reassessment of volume status is key. If there is uncertainty with volume status in the setting of worsening kidney function or suspicion for low cardiac output, a right heart catheterization is performed to help inform management. Persistent congestion warrants the institution of other strategies such as sequential nephron blockade or even ultrafiltration if needed. The presence of low cardiac output can be addressed by treating underlying causes such as active ischemia and treatment with inotropes and vasodilators depending on the hemodynamic status and mechanical circulatory support if necessary. Abbreviations: AV, atrioventricular; CRT, cardiac resynchronization therapy; ENaC, epithelial sodium channel; GDMT, guideline-directed medical therapy; IVC, inferior vena cava; LV, left ventricle; NSAID, nonsteroidal anti-inflammatory drug; OAT, organic anion transporter; POCUS, point-of-care ultrasound; pulm HTN, pulmonary hypertension; RV, right ventricle; VEXUS, Venous Excess Ultrasound score.

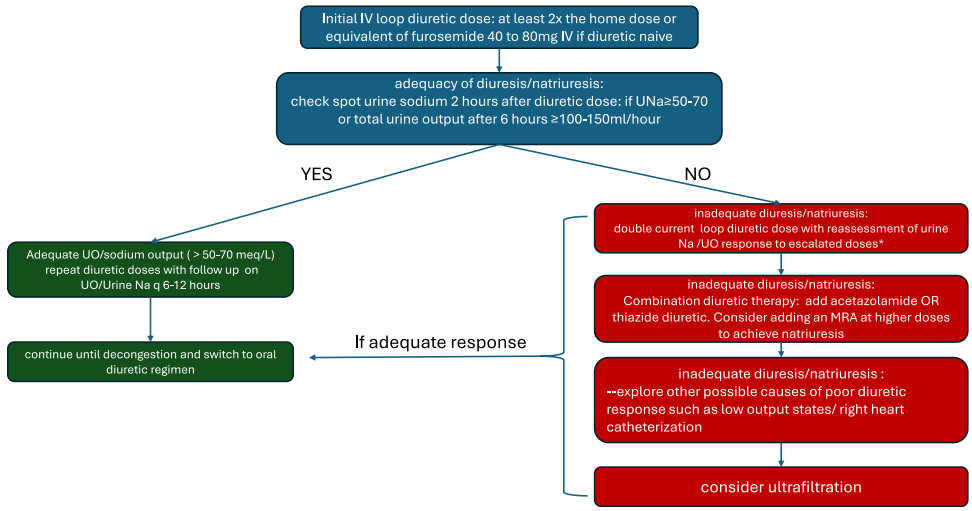


Figure 4. Algorithm for diuretic escalation and assessment of diuretic response. IV, intravenous; MRA, mineralocorticoid receptor antagonist; UNa, urine sodium; UO, urine output.

GDMT Optimization in HF and Kidney Disease

Case 6: *A 63-year-old woman with HFrEF (EF, 30%) and CKD (stage G3a A1) is hospitalized with volume overload with worsening kidney function. During admission, she received loop diuretic agents, her kidney function returned to preadmission measurements, and she regained her dry weight of 215 lbs. Before hospitalization, her only medication was carvedilol, and she was discharged with a prescription for furosemide 40 mg once daily.*

Question 7: What additional classes of agents constitute GDMT for this individual?

(a) Angiotensin receptor/neprilysin (ARN) inhibitors, sodium/glucose cotransporter 2 (SGLT2) inhibitors
(b) ARN inhibitors, SGLT2 inhibitors, mineralocorticoid receptor antagonists (MRAs)
(c) ARN inhibitors, MRAs

For the answer to this question, see the following text.

The treatment and management of HF (with a reduced, mildly reduced, or preserved EF) has evolved significantly over time through numerous trials. Based on current data, GDMT for HFrEF involves the use of the following classes of medication: β-adrenergic receptor blockers, RAAS inhibitors (angiotensin-converting enzyme inhibitors, ARBs, or ARN inhibitors), SGLT2 inhibitors, and MRAs; therefore, (b) is the best answer to Question 7. These medication classes constitute the "pillars of therapy" in HFrEF, and, when used in tandem and titrated quickly to reach target goals, have a mortality benefit, along with reduced hospitalizations for HF, as recently demonstrated in the pragmatic STRONG HF trial.

β-Adrenergic receptor blockers in HFrEF are used widely across the range of CKD stages with the least therapeutic inertia, likely because they are eGFR-neutral. Angiotensin-converting enzyme inhibitors and ARBs have been shown to have mortality and morbidity benefits mainly in the HFrEF (EF <40%) population, as seen in the previous landmark trials (CONSENSUS, SOLVD, Val-HeFT; Table 2); however, they tend to be underused in HFrEF with CKD because of concerns for hyperkalemia and acute kidney injury. The ARN inhibitor sacubitril/valsartan has demonstrated important clinical benefits in cardiovascular and/or kidney outcomes in patients with HFrEF and HF with a mildly reduced EF. Neprilysin is an endopeptidase that cleaves a variety of peptides such as natriuretic peptides, bradykinin, adrenomedullin, substance P, angiotensin I and II, and endothelin. It has a broad role in several cardiovascular, kidney, pulmonary, gastrointestinal, endocrine, and neurologic functions. The combined ARN inhibitor has been developed with an intent to increase vasodilatory natriuretic peptides and prevent counterregulatory activation of the angiotensin system. When comparing the use of ARN inhibitors versus ARBs for HF with preserved EF, the PARAGON-HF study (Prospective Comparison of ARNI with ARB Global Outcomes in HF with preserved EF) found that sacubitril/valsartan was associated with a lower eGFR slope decline and decreased HF-related hospitalizations compared with valsartan alone. Sacubitril/valsartan also demonstrated reduced cardiovascular and all-cause mortality in HFrEF versus enalapril in the PARADIGM-HF trial (Prospective Comparison of ARNI with ACE-I to Determine Impact on Global Mortality and Morbidity in Heart Failure).

Although ARN inhibitors have shown benefits in slowing eGFR slope decline in HF, they are associated with increased albuminuria. Possible explanations for this paradoxical and opposing directionality of increased albuminuria and reduced eGFR decline may include a preferential vasodilation of the afferent arteriole and a relative vasoconstriction of the efferent arteriole in the context of lower blood

Table 2. Summary Table on Key Clinical Trials on Guideline Directed Medical Therapy for Heart Failure

Study, Year	Drug	Control	Population Enrolled	CV Outcome	*P* Value	Renal Function Exclusion
ARNIs						
PIONEER-HF, 2021	Sacubitril/valsartan	Enalapril	HFrEF	Time-averaged proportional change in NT-proBNP concentration from baseline through weeks 4 and 8: −46.7% vs. −25.3%	<0.001	–
PARADIGM-HF, 2014	Sacubitril/valsartan	Enalapril	HFrEF	CV mortality or HF hospitalization: 21.8% vs. 26.5%	<0.001	GFR <30
PARAMOUNT, 2012	Sacubitril/ valsartan (LC696)	Valsartan	HFrPEF	Change in NT-proBNP at 12 weeks vs valsartan; LCZ696/valsartan ratio, 0.77	0.005	GFR <30
β-Blockers						
MERIT-HF, 1999	Metoprolol	Placebo	HFrEF	All-cause mortality: 0.072 vs 0.11 per patient-year	<0.001	–
COPERNICUS, 2002	Carvedilol	Placebo	HFrEF	Annual mortality: 12.8% vs 19.7%	<0.001	–
MRA						
RALES, 1999	Spironolactone	Placebo	HFrEF	All-cause mortality: 35% vs 46%	<0.001	Cr >2.5
EMPHASIS-HF, 2011	Eplerenone	Placebo	HFrEF	CV death or HF hospitalization: 18.3% vs 25.9%	<0.001	GFR <30
EPHESUS, 2003	Eplerenone	Placebo	HFrEF	All-cause mortality: 19.8% vs 24.7%	0.008	Cr > 2.5
TOPCAT, 2014	Spironolactone	Placebo	HFpEF	CV mortality, aborted cardiac arrest, or HF hospitalization: 18.6% vs 20.4%	0.1	Cr >2.5 or GFR <30
Nonsteroidal MRAs						
FIDELIO-DKD, 2021	Finerenone	Placebo	NA	CV mortality, nonfatal MI, nonfatal stroke, or HF hospitalization[a]: 13.0% vs 14.8%	0.03	–
FIGARO-DKD, 2021	Finerenone	Placebo	NA	CV death, MI, stroke, hospitalization for HF: 12.4% vs 14.2%	0.03	GFR <25
FINEARTS-HF, 2024	Finerenone	Placebo	HFmrEF, HFpEF	Composite of worsening HF events (first or recurrent unplanned HF hospitalization or urgent visit) and CV death (14.9 vs 17.7 events per 100 patient-years)	0.007	GFR <25, serum K >5.0
ARBs						
Val HeFT, 2001	Valsartan	Placebo	HFrEF	All-cause mortality: 19.7% vs 19.4%	0.8	Cr >3.4
ACE inhibitors						
CONSENSUS, 1987	Enalapril	Placebo	NYHA class IV	6-month mortality: 26% vs 44%	0.002	Cr >3.4
SOLVD, 1991	Enalapril	Placebo	HFrEF	All-cause mortality: 35% vs 40%	0.007	Cr >2.0
SGLT2 inhibitors						
DAPA-HF, 2019	Dapagliflozin	Placebo	HFrEF	Worsening HF (hospitalization or urgent visit resulting in IV therapy for HF) or CV mortality: 16.3% vs 21.2%	<0.001	GFR <30
DELIVER, 2022	Dapagliflozin	Placebo	EF >40%	HF hospitalization, urgent HF visit, or CVD mortality: 16.4% vs 19.5%	<0.001	GFR <25
EMPEROR-PRESERVED, 2015	Empagliflozin	Placebo	EF > 40%	Death from cardiovascular causes or hospitalization for HF: 13.8% vs 17.1%	<0.001	GFR <20

(Continued)

Table 2 (Cont'd). Summary Table on Key Clinical Trials on Guideline Directed Medical Therapy for Heart Failure

Study, Year	Drug	Control	Population Enrolled	CV Outcome	*P* Value	Renal Function Exclusion
EMPEROR-REDUCED, 2020	Empagliflozin	Placebo	HFrEF and NT-ProBNP >600	Composite of cardiovascular death and first hospitalization for decompensated HF: 19.4% vs 24.7%	<0.001	GFR <20
SOLOIST-WHF, 2023	Sotagliflozin	Placebo	HF (all)	Total CV death, hospitalization for HF, or urgent visit for HF: 70 vs 98 events per 100 patient-years	<0.001	GFR <30
EMPULSE, 2022	Empagliflozin	Placebo	HF (all)	Composite of death, number of HF events, time to first HF event, and change in KCCQ Total Symptom Score: clinical benefit, 53.9% vs 39.7%	0.005	NA

Abbreviations: ACE, angiotensin-converting enzyme; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor blockers/neprilysin inhibitor; Cr, creatinine; CV, cardiovascular; GFR, glomerular filtration rate; HF, heart failure; HFmrEF, heart failure with mildly reduced ejection fraction; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; IV, intravenous; KCCQ, Kansas City Cardiomyopathy Questionnaire; MI, myocardial infarction; MRA, mineralocorticoid receptor antagonist; NA, not applicable; NT-proBNP, N-terminal pro–brain natriuretic peptide; NYHA, New York Heart Association; SGL2, sodium/glucose cotransporter 2.
[a]Secondary outcome; primary outcome was kidney failure.

pressure, as well as the enhanced bioavailability of natriuretic peptides with ARN inhibitor use.

Steroidal MRAs are also among the 4 pillars of GDMT in HFrEF but remain underused, especially in patients with advanced CKD. The landmark trials RALES (spironolactone in patients with HFrEF) and EPHESUS (left ventricular dysfunction in the setting of recent myocardial infarction) showed reduced risk for cardiovascular death and hospitalizations in this patient population. However, among patients with preserved EF, spironolactone was not shown to be beneficial in improving clinical outcomes, as had been seen in the TOPCAT trial. A post hoc analysis suggested potential benefits of spironolactone based on geographic regions of enrollment, but true clinical effects remained speculative. In this context, the nonsteroidal MRA finerenone demonstrated significant reduction in the primary end point of total HF events and cardiovascular death in patients with mildly reduced and preserved EF in the FINEARTS-HF trial. Although steroidal and nonsteroidal MRAs influence several overlapping pathways in HF and kidney disease progression, finerenone has unique physicochemical properties that distinguish it from steroidal MRAs. These include equal distribution between heart and kidney tissues, a shorter half-life, higher MR selectivity and affinity, and lower rates of hyperkalemia. Patient-level pooled data across pivotal studies of finerenone across a broad range of cardiovascular, kidney, and metabolic conditions (FIGARO-DKD, FIDELIO-DKD, and FINEARTS-HF) demonstrate important benefits with prespecified kidney composite outcomes, HF hospitalizations, and a signal for reduction in all-cause death.

The class of SGLT2 inhibitor agents have been shown to have multiple cardiac/kidney protective effects through various mechanisms independent of glycemic control or EF and across a broad range of eGFRs. They encompass improved cardiac metabolism, reduced oxidative stress, modulation of neurohormonal pathways, attenuation of myocardial inflammation, and preservation of endothelial function. SGLT2 inhibitors also function as diuretic augmenters, as shown in the EMPULSE trial, and are attractive options for enhanced decongestion and GDMT in the hospitalized patient with HF. SGLT2 inhibitors are generally safely initiated for patients with an eGFR ≥20 mL/min/1.73 m^2; as kidney function progressively worsens, SGLT2 inhibitors can be continued until the initiation of dialysis and/or renal replacement therapy (Figs 5 and 6; Table 2). In relation to this, the use of glucagon-like peptide receptor agonists has also recently garnered attention, especially in cases of HF with preserved EF with the obesity phenotype. Recent trial evidence has also showed benefits in reduction of kidney events and cardiovascular death in patients with type 2 diabetes and CKD.

Implementation of GDMT in patients with HF and CKD is suboptimal. This may be a result of the practice of "renalism" to avoid the largely perceived risk of worsening kidney function. RAAS inhibitors and MRAs are particularly underused in patients with CKD as a result of

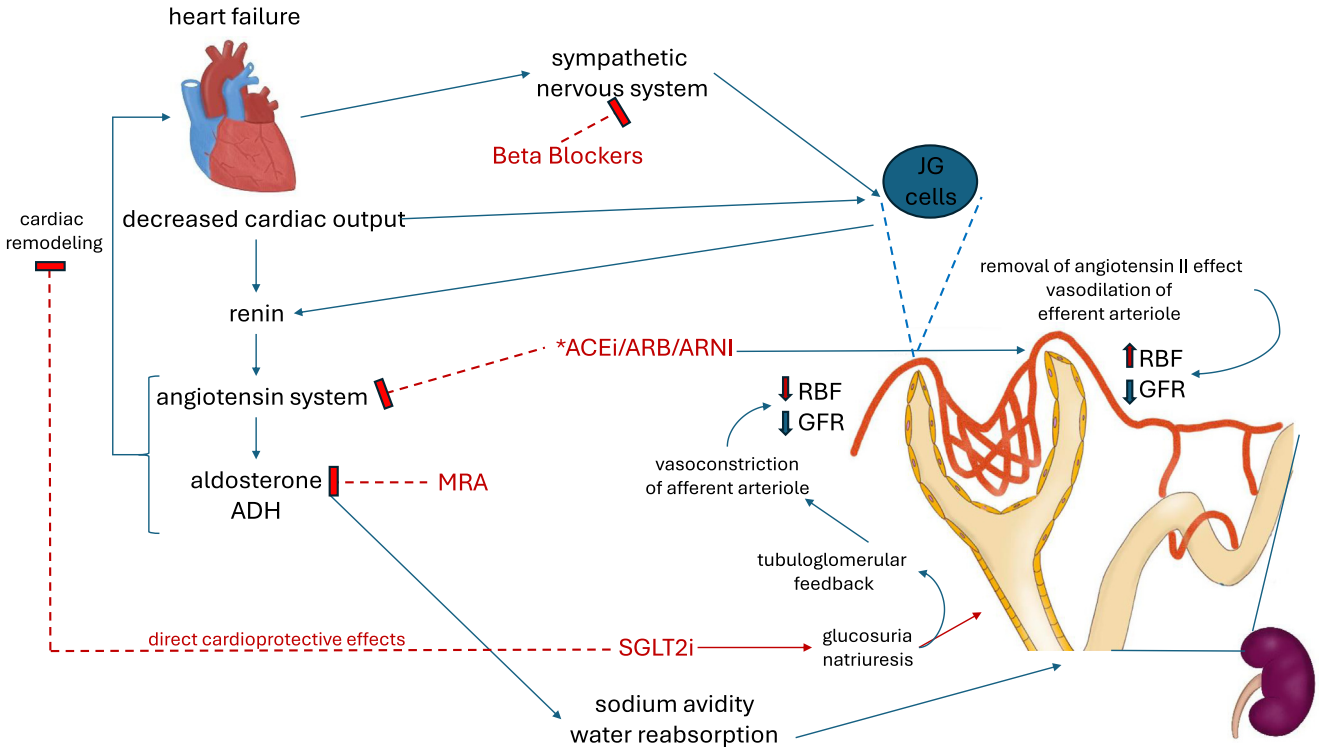


Figure 5. Summary of cardiac and kidney mechanisms of guideline-directed medical therapy for heart failure. Heart failure induces a state of maladaptive neurohormonal response to a decreased effective circulating volume. The renin-angiotensin-aldosterone system with antidiuretic hormone is activated and is at the heart of heart/kidney interactions. Angiotensin II leads to preferential vasoconstriction of the efferent renal arterioles, reducing kidney blood flow but increasing glomerular filtration rate. Aldosterone and antidiuretic hormone lead to increased sodium and water reabsorption, perpetuating the cycle of congestion. Renin-angiotensin-aldosterone system inhibitors such as angiotensin-converting enzyme/angiotensin receptor blockers and angiotensin receptor blockers/neprilysin inhibitors antagonize angiotensin II, causing efferent vasodilation, leading to higher renal blood flow, but can lower glomerular filtration rate transiently. Together with mineralocorticoid receptor antagonism, β-blockers have an effect on the sympathetic nervous system and renin secretion, which can lead to long-term reduction in harmful downstream effects of fibrosis and cardiac remodeling. Sodium/glucose cotransporter 2 inhibitors lead to glucosuria and natriuresis by inhibiting proximal tubular reabsorption. This leads to increased sodium delivery to the distal tubule, causing tubuloglomerular feedback that also causes vasoconstriction. This is one of the implicated mechanisms of kidney protection by the prevention of glomerular hyperfiltration, especially in diabetic kidney disease. However, there are also studies showing effects of sodium/glucose cotransporter 2 inhibitors on vasodilation of the efferent arteriole also increasing kidney blood flow. In addition, the use of sodium/glucose cotransporter 2 inhibitors in addition to loop diuretic agents has led to increased rates of decongestion and diuretic efficacy in various studies. Sodium/glucose cotransporter 2 inhibitors also shift the metabolic profile of heart muscle toward more ketone use, which has less oxygen demand. This mechanism is implicated in its direct cardioprotective effects. Abbreviations: ACEi, angiotensin-converting enzyme inhibitor; ADH, antidiuretic hormone; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor blockers/neprilysin inhibitor; GFR, glomerular filtration rate; JG, juxtaglomerular cells; MRA, mineralocorticoid receptor antagonist; RBF, renal blood flow; SGLT2i, sodium/glucose cotransporter 2 inhibtor.

concerns of hyperkalemia and worsening kidney function. The STRONG-HF study (Safety, Tolerability, and Efficacy of Rapid Optimization, Helped by NT-proBNP Testing of Heart Failure Therapies) investigated the implementation of quadruple GDMT in HFrEF comparing the standard approach versus high-intensity uptitration (ie, 2 weeks). The results showed a reduced risk of death or hospitalization for HF at 180 days with rapid implementation and uptitration of GDMT in a real-world population, with an acceptable safety profile. However, it should be noted that this trial excluded patients with an eGFR <30 mL/min/1.73 m^2 and those who had a clear intolerance to high doses of β-blockers, ACE inhibitors, or ARBs, which can be common in the subset of patients with advanced CKD.

Additional Readings

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HFrEF therapies	CKD 1 and 2	CKD 3	CKD 4	CKD 5
Beta Blocker	Strong	Strong	Limited	Absent
MRA	Strong	Strong	Limited	Absent
Non-steroidal MRA	Strong	Strong	Strong (up to eGFR> 25 cc/min)	Absent
ARNi	Strong	Strong	Limited	Absent
ACEi/ARB	Strong	Strong	Limited	Absent
Diuretics	Absent	Absent	Absent	Absent
SGLT2i	Strong	Strong	Strong (eGFR> 20 cc/min)	Limited

Figure 6. Summary of the levels of evidence for cardiac medications according to chronic kidney disease stage. Abbreviations: ACEi, angiotensin-converting enzyme inhibitor; ARB, angiotensin receptor blocker; ARNI, angiotensin receptor blocker/neprilysin inhibitor; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate; MRA, mineralocorticoid receptor antagonist; SGL2, sodium/glucose cotransporter 2.

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Challenges in GDMT Optimization Especially in CKD

Case 7: *A 65-year-old man with a history of HFrEF (EF, 35%), diabetes, hypertension, and CKD (stage G3a A3) is seen in the outpatient clinic. His home medications include furosemide 40 mg twice daily and metoprolol succinate and lisinopril 20 mg. His temperature is 36.1°C, pulse rate is 88 bpm, respiration rate is 15 breaths/min, and blood pressure is 100/57 mm Hg. Physical examination discloses no jugular venous distension, with clear bilateral breath sounds and no audible crackles. His laboratory studies show a sodium concentration of 136 mEq/L, potassium concentration of 5.3 mEq/L, chloride concentration of 102 mEq/L, serum creatinine concentration of 1.6 mg/dL (at baseline), SUN concentration of 15 mg/dL, and NT-proBNP concentration of 1,500 pg/mL. Treatment with lisinopril was stopped, and the patient was referred to the nephrology service to help with GDMT optimization.*

Question 8: What is the next best step for this patient?

(a) Start spironolactone
(b) Increase the furosemide dose
(c) Add an SGLT2 inhibitor together with lisinopril
(d) Maintain deescalation of lisinopril
(e) Add metolazone 3 times per week

For the answer to this question, see the following text.

Hyperkalemia is commonly one of the limiting adverse effects encountered when titrating GDMT in patients with HF and CKD. Despite the limitations of such observational data, the fact that GDMT withdrawal is associated with poorer clinical outcomes suggests that we should try hard to not stop or to quickly restart GDMT whenever possible in patients with HF and CKD. Various strategies can be considered in the approach to hyperkalemia in HF and CKD. In a meta-analysis of randomized controlled trials in patients with type 2 diabetes with high cardiovascular risk and/or CKD, SGLT2 inhibitors reduced the risk of serious hyperkalemia (hazard ratio, 0.84; 95% CI, 0.76–0.93). This was true across subgroups of patients taking RAAS inhibitors and MRAs. This becomes an opportunity for simultaneous introduction/reintroduction of further components of GDMT to help improve not only cardiovascular but also kidney outcomes in these patients; therefore, (c) is the best answer to Question 8. Another alternative is switching to an ARN inhibitor. In an analysis of the PARADIGM-HF trial, among patients with symptomatic HFrEF treated with MRAs, severe hyperkalemia was more common with enalapril treatment compared with sacubitril/valsartan (hazard ratio, 1.37; 95% CI, 1.06-1.76). Potassium binder use can also be considered for GDMT maintenance in HF. In the DIAMOND trial, among patients with HFrEF and hyperkalemia or a history of hyperkalemia, randomization to patiromer concurrently with high-dose RAAS inhibitor therapy was associated with lower rates of hyperkalemia (K concentration >5.5 mmol/L) compared with placebo (hazard ratio, 0.63; 95% CI, 0.45-0.87). Finally, a simple rechallenge even with the same or a lower dose remains a feasible option. In the DIAMOND trial, >80% of patients in the placebo group did not go on to

develop hyperkalemia despite the absence of patiromer use with concomitant uptitration of RAAS inhibitors. This suggests that, at times, a careful rechallenge can be attempted with close monitoring.

Although increasing or adjusting diuretic agent doses can help with management of the patient's potassium levels, diuretic agents should be titrated to a dose that would best maintain euvolemia in these patients, and this patient was relatively euvolemic on presentation. Introduction of SGLT2 inhibitors (and, as recently reported, also glucagon-like peptide-1 receptor agonists) may lead to a reduced need for loop diuretic agent doses and lower rates of hyperkalemia.

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Case 8: *A 45-year-old woman with a history of HFrEF (EF, 35%), diabetes, and hypertension is seen in the outpatient clinic. She has a new diagnosis of HF and had been hospitalized 1 week earlier. Her home medications include furosemide 40 mg/d and metoprolol succinate. She had been prescribed lisinopril 20 mg/d and empagliflozin 10 mg/d 2 weeks earlier. She feels well and is euvolemic on examination. Her latest laboratory studies show a sodium concentration of 136 mEq/L, potassium concentration of 4.5 mEq/L, chloride concentration of 102 mEq/L, serum creatinine concentration of 1.5 mg/dL (increased from 1.2 mg/dL 2 weeks earlier), SUN concentration of 16 mg/dL, and NT-proBNP concentration of 900 pg/mL (decreased from 1,400 pg/mL 2 weeks earlier).*

Question 9: What is the next best step for this patient?

(a) Continue lisinopril and empagliflozin
(b) Stop lisinopril
(c) Switch lisinopril to valsartan
(d) Stop empagliflozin

Answer: For the answer to the question, see the following text.

Changes in kidney function in the setting of HF management have to be interpreted in the context of decongestion and GDMT initiation/uptitration. Worsening kidney function, especially in the setting of aggressive decongestion, is frequently encountered. However, these changes in GFR may not reflect true tubular injury (Fig 7). In fact, in the setting of acute decompensated HF, worsening kidney function in the setting of successful decongestion is associated with lower mortality and reduced hospitalization rates than failure to decongest in the setting of stable or improving kidney function. Several caveats exist in interpreting changes in markers of glomerular function (especially serum creatinine) in the acute HF state and in patients with sarcopenia with HF. This makes serum creatinine particularly problematic for an accurate assessment of eGFR in patients with HF. Therefore, moving away from a siloed interpretation of fluctuating markers of glomerular filtration for prognostication toward recognizing persistent congestion and deploying a targeted approach to treating diuretic resistance may help improve outcomes in patients with HF. This is an area for future research in clinical trials. Initiation of GDMT for HF can also contribute to fluctuations in kidney function. Initiation of a RAAS inhibitor or SGLT2 inhibitor can be associated with fluctuations in serum creatinine of as much as 0.3 mg/dL or more within the first 2 weeks of initiation. Guidelines have recommended continuation if the eGFR decrease is <30% versus baseline. In a prespecified analysis of the EMPA-KIDNEY trial, although there was an approximately 6% decrease in eGFR in the first 2 months, over time, the chronic slope of eGFR decline was halved in the empagliflozin group compared with the placebo group, showing slowing of kidney disease progression with a relative difference of 50% (95% CI, 42%-58%). Continued use of GDMT in patients with HF and CKD is associated with improvement in cardiac and kidney outcomes, and the definition of stage 1 acute kidney injury by standard criteria do not apply to the hemodynamic context of decongestion and GDMT use in HF; therefore, (a) is the best answer to Question 9. It is important to carefully ascertain if there is true tubular injury in the appropriate clinical context (eg, sepsis, bleeding), and urine microscopy is a valuable tool in this regard, with excellent negative predictive value for a diagnosis of clinically important intrinsic kidney injury.

Additional Readings

- KDIGO 2020 Clinical practice guideline for diabetes management in chronic kidney disease. *Kidney Int.* 2020;98(4 suppl):S1-S115. doi:10.1016/j.kint.2020.06.019
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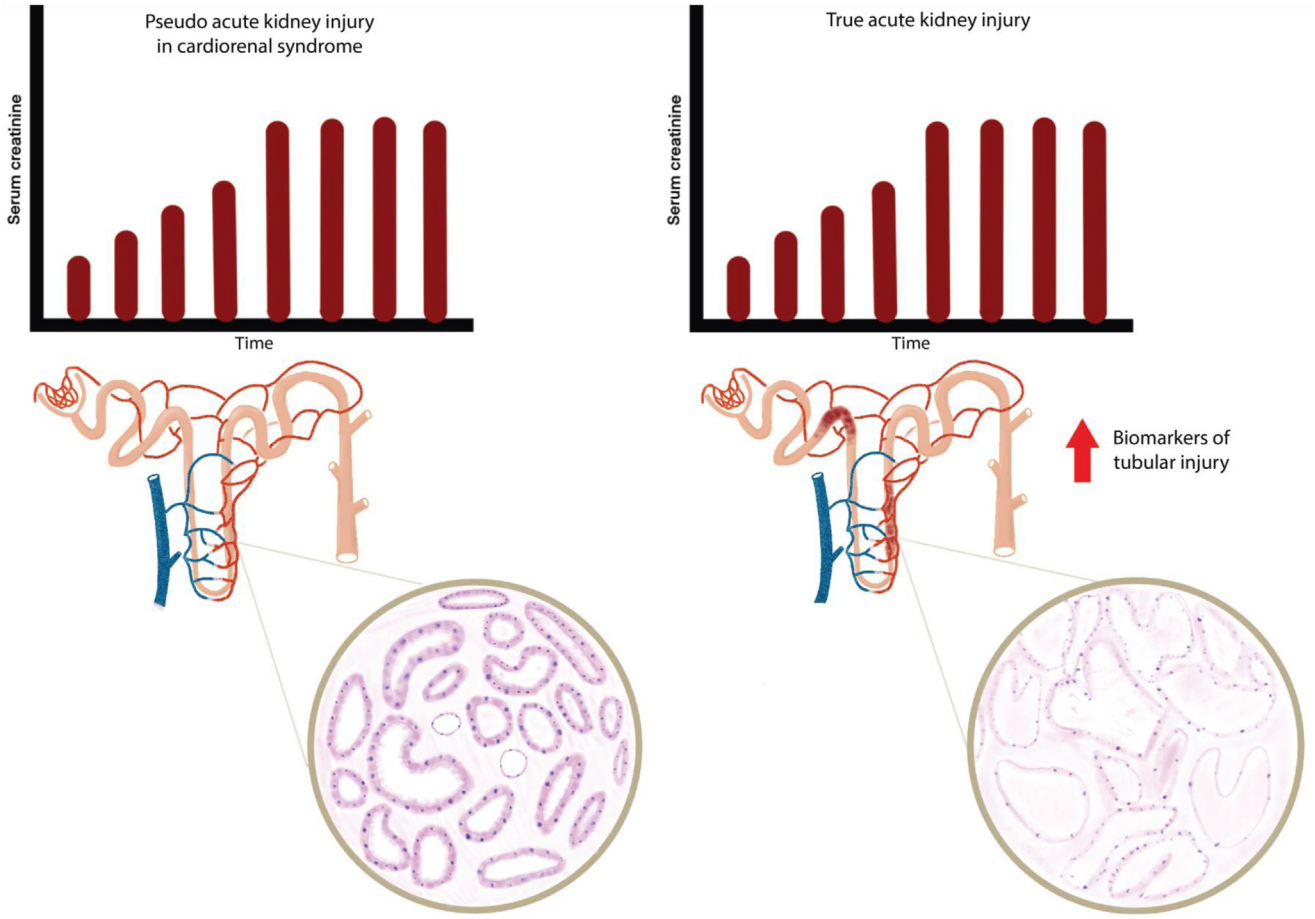


Figure 7. Worsening kidney function in heart failure. This figure presents 2 different clinical scenarios in which serum creatinine increase and perceived worsening renal function may occur. On the left, there is a serum creatinine increase but no actual tubular damage. This can happen in the setting of decongestion, cardiorenal syndrome, and even renin-angiotensin-aldosterone system inhibitor use. On the right, increased serum creatinine level happens in the setting of acute tubular necrosis, and novel biomarkers such as NGAL (neutrophil gelatinase-associated lipocalin) and KIM-1 (Kidney Injury Molecule-1) can be of value to help differentiate true tubular damage.

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